

QUESTIONS

Q1

Palindrome number/string
10 marks

Q2

First n perfect numbers
10 marks

Q3

Decimal number equivalent to
binary number and octal
number
10 marks

Q4

Nth max number and nth min
number in array then sum of it
and difference of it
10 marks

Q5

Find the average positive and
negative numbers upto -1
10 marks

Q6

Removing duplicates in array
10 marks

Q7

Matrix multiplication
10 marks

Q1 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```
1 import java.util.Scanner;
2 public class PalindromeNumber{
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         int num,temp,rev=0,rem;
7         System.out.println("Enter a number:");
8         num=sc.nextInt();
9         temp=num;
10        while(num!=0)
11        {
12            rem=num%10;
13            rev=rev*10+rem;
14            num=num/10;
15        }
16        if(temp==rev)
17            System.out.println("Palindrome");
18        else
19            System.out.println("Not a Palindrome");
20        sc.close();
21    }
22 }
```

Run

Save

Input

121

Output

Enter a number:
Palindrome

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Q2 First n perfect numbers

10 marks

First n perfect numbers

Your Code

```
1 import java.util.Scanner;
2 public class PerfectNumber
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         System.out.println("Enter N:");
8         int n= sc.nextInt();
9         int []arr=new int[n];
10        int count=0;
11        int num=2;
12        while (count<n)
13        {
14            int sum=0;
15            for(int i=1;i<num;i++)
16            {
17                if(num%i==0)
18                    sum+=i;
19            }
20            if(sum==num)
21            {
22                arr[count]=num;
23                count++;
24            }
25        }
26    }
27 }
```

Input

3

Output

```
Enter N:
First3Perfect Number:
6
28
496
```


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10 marks



Q2

First n perfect numbers

10 marks



Q3

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10 marks



Q4

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number in array then sum of it
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Q5

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Q6

Removing duplicates in array

10 marks



Q7

Matrix multiplication

10 marks



Q3 Decimal number equivalent to binary number and octal number

10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 import java.util.Scanner;
2 public class DecimalToBinaryOctal {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int decimal, i = 0;
6         int[] binary = new int[32];
7         int[] octal = new int[32];
8         System.out.print("Enter a decimal number: ");
9         decimal = sc.nextInt();
10        int temp = decimal;
11        while (temp > 0) {
12            binary[i] = temp % 2;
13            temp = temp / 2;
14            i++;
15        }
16        System.out.print("Binary = ");
17        for (int j = i - 1; j >= 0; j--) {
18            System.out.print(binary[j]);
19        }
20        temp = decimal;
21        i = 0;
22        while (temp > 0) {
23            octal[i] = temp % 8;
24            temp = temp / 8;
```

Input

25

Output

```
Enter a decimal number: Binary =
11001
Octal = 31
```

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number in array then sum of it
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Find the average positive and
negative numbers upto -1
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Matrix multiplication
10 marks

Q4 Nth max number and nth min number in array then sum of it and difference of it

10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.Arrays;
2 import java.util.Scanner;
3 public class NthMaxMin {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print("Enter array size: ");
7         int n = sc.nextInt();
8         int[] arr = new int[n];
9         System.out.println("Enter array elements:");
10        for (int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13        Arrays.sort(arr);
14        System.out.print("Enter N: ");
15        int k = sc.nextInt();
16        int nthMin = arr[k - 1];
17        int nthMax = arr[n - k];
18        int sum = nthMin + nthMax;
19        int diff = nthMax - nthMin;
20        System.out.println("Nth Minimum = " + nthMin);
21        System.out.println("Nth Maximum = " + nthMax);
22        System.out.println("Sum = " + sum);
23        System.out.println("Difference = " + diff);
24        sc.close();
25    }
26 }
```

Input

```
5
10 20 30 40 50
3
```

Output

```
Enter array size: Enter array
elements:
Enter N: Nth Minimum = 30
Nth Maximum = 30
Sum = 60
Difference = 0
```


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Q6 Removing duplicates in array

10 marks

Removing duplicates in array

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter number of elements: ");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("Enter elements: ");
9         for (int i = 0; i < n; i++) {
10             arr[i] = sc.nextInt();
11         }
12         int[] temp = new int[n];
13         int j = 0;
14         for (int i = 0; i < n; i++) {
15             boolean isDuplicate = false;
16             for (int k = 0; k < j; k++) {
17                 if (arr[i] == temp[k]) {
18                     isDuplicate = true;
19                     break;
20                 }
21             }
22             if (!isDuplicate) {
23                 temp[j] = arr[i];
24                 j++;
25             }
26         }
27         System.out.println("Array after removing duplicates:");
28         for (int i = 0; i < j; i++) {
29             System.out.print(temp[i] + " ");
30         }
31     }
32 }
```

Input

```
7
1 2 2 3 4 4 5
```

Output

```
Enter number of elements: Enter
elements:
Array after removing duplicates:
1 2 3 4 5
```

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Matrix multiplication
10 marks

Q7 Matrix multiplication

10 marks

Matrix multiplication

Your Code

```
1 import java.util.Scanner;
2 public class MatrixMultiplication
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int[][]a=new int[2][2];
8         int[][]b=new int[2][2];
9         int[][]c=new int[2][2];
10        System.out.println("Enter first 2x2 matrix;");
11        for(int i=0;i<2;i++)
12        {
13            for(int j=0;j<2;j++)
14            {
15                a[i][j]=sc.nextInt();
16            }
17        }
18        System.out.println("Enter second 2x2 matrix:");
19        for(int i=0;i<2;i++)
20        {
21            for(int j=0;j<2;j++)
22            {
23                b[i][j]=sc.nextInt();
24            }
25        }
26    }
27 }
```

Input

```
1 2
3 4
5 6
7 8
```

Output

```
Enter first 2x2 matrix;
Enter second 2x2 matrix:
Result Matrix:
19 22
43 50
```


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Q5

Find the average positive and
negative numbers upto -1
10 marks

Q6

Removing duplicates in array
10 marks

Q7

Matrix multiplication
10 marks

Q5 Find the average positive and negative numbers upto -1

10 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int posCount = 0, negCount = 0;
6         int posSum = 0, negSum = 0;
7         System.out.println("Enter numbers (enter -1 to stop):");
8         while (true) {
9             int num = sc.nextInt();
10            if (num == -1) {
11                break;
12            }
13            if (num > 0) {
14                posSum += num;
15                posCount++;
16            } else if (num < 0) {
17                negSum += num;
18                negCount++;
19            }
20        }
21        double posAverage = (posCount != 0) ? (double) posSum / posCount : 0;
22        double negAverage = (negCount != 0) ? (double) negSum / negCount : 0;
23        System.out.println("Average of positive numbers: " + posAverage);
24        System.out.println("Average of negative numbers: " + negAverage);
25    }
26 }
```

Input

5 -3 8 -2 10 -7 -1

Output

```
Enter numbers (enter -1 to stop):
Average of positive numbers:
7.666666666666667
Average of negative numbers: -4.0
```